CO1 AT3

Arunprasath R
192524077
CSA0711
AI
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1) Depth Limited Search :-

Level-4

→ Definition and Data :-

- Depth Limited Search is defined as the Uniformed Search that Variation of DFS that restrict search to specific maximum depth.
- It prevent search from going indefinitely deep.

→ Strategy and Data Structure :-

- Strategy : Expand deepest node but do not expand nodes at depth.
- Data structure : Stack or recursive calls

→ Algorithm :-

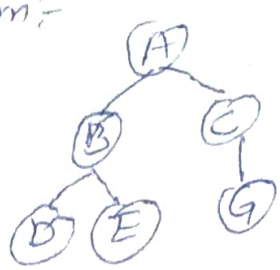
- Set depth limit to 0.
- Then perform the uniformed search
- Rectify the solution of depth limit.
- Return & Repeat.

→ pseudocode :-

→

```
DLS (node, goal, limit)
  if node is goal;
    return solution
  if limit == 0
    return cutoff
  for each child of node
    result = DLS (child, goal, limit - 1)
  if result is soln;
    return result
  return failure or cutoff.
```

→ Diagram :-



Expansion order : A → B → C

→ properties :-

- completeness : NO, if $l < d$
- optimality : NO
- Time complexity : $O(b^e)$
- Space complexity : $O(bl)$

→ Advantage :-

- prevent unlimited exploration & control memory usage.

→ Disadvantage :-

- may miss a solution deeper than limit.

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2) Iterative Deepening Search (IDS) :-

→ Definition & Data :-

- IDS is defined as the combine advantage of DFS and BFS.
- It repeatedly perform DLS increase depth limit.

→ Strategy & Data structure :-

- Strategy : It perform DLS repeatedly with limit.
- D.S : Stack or recursion.

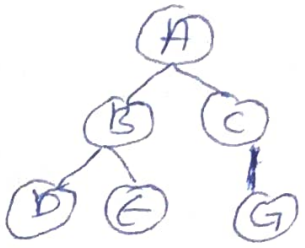
→ Algorithm :-

- set depth limit to 0
- perform DLS
- If goal found, return solution
- Repeat

Pseudocode :-

```
IDS (start, goal)
    for limit 0 to infinity;
        result = DLS (start, goal, limit)
    if result is solution
        return result.
```

→ Diagram :-



order :- A → C → G

A → B → D → E → C → G.

→ Advantage :-

- IDS also helps to manage unlimit iterative deepening and control usage.

→ Disadvantage :-

- IDS may have a chance of dismiss a solution deeper than its accurate.

→ Peroperty :-

- completeness : Yes
- optimality : Yes
- Time complexity : $O(b^d)$
- Space complexity : $O(bd)$.